

■ ■ Nashik Roads & Civic Monitor: Deliverable 3 — Issue / Bug Log

Submission Requirement: Table of Issues/Errors Identified & Debugging/Fixes | **Format:** Table: Issue ID, Description, Severity, Root Cause, Fix, Status, Date

Standard: IEEE 1044 Anomaly Classification | **Evaluation Status:** 14 Issues Logged / 14 Resolved (100% Fix Rate) | **Target:** Nashik Municipal Corporation

TOTAL ISSUES LOGGED 14	CRITICAL (BLOCKER) 3 (Resolved)	HIGH SEVERITY 4 (Resolved)	MEDIUM SEVERITY 4 (Resolved)	RESOLUTION RATE 100% (14/14)
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Issue ID	Description	Severity	Root Cause	Fix Implemented	Status	Date
BUG-001	GIS Map runtime error "Map container is already initialized" during tab switching	Critical	Leaflet L.map() attached to DOM without unmounting prior map instance during React re-renders	Added cleanup return function inside useEffect hook calling mapInstance.remove() and resetting DOM ref	Resolved	2026-09-12
BUG-002	Newly submitted citizen grievances failed to display as pins on interactive Leaflet GIS map	Critical	GIS map was rendering only static projects array and had not subscribed to live tickets state	Integrated useCivicStore tickets listener creating dynamic Leaflet circle markers with hazard colors	Resolved	2026-09-15
BUG-003	Grievance form accepted invalid phone numbers (e.g. "123" or alphabet characters) without error	High	Mobile input field only verified non-empty string and did not enforce Indian 10-digit regex	Added client-side regex check /^[6-9]\d{9}\$/ in ComplaintFormModal and validated payload schema in Express	Resolved	2026-09-18
BUG-004	Landmark selection defaulted to generic ward centroid coordinates rather than exact road location	High	Absence of localized landmark coordinate snapping; form relied on fallback ward centroid	Implemented LANDMARK_COORDINATES lookup table with fuzzy matching plus interactive mini-map pinpoint picker	Resolved	2026-09-20
BUG-005	Unauthorized users were able to mark contractor road works as "Resolved" without verification	Critical	Missing role authorization and reporter-identity verification guards on ticket resolution actions	Implemented reporter-only verification logic requiring matching citizen ID / admin passkey before state transition	Resolved	2026-09-22
BUG-006	Uploading 5+ MB mobile camera photos caused server memory spikes and excessive database JSON payload size	High	Raw JPEG files were directly encoded to uncompressed Base64 strings (7+ MB per ticket)	Implemented client-side HTML5 Canvas 2D image downscaling and WebP compression pipeline (<250KB payload)	Resolved	2026-09-24
BUG-007	Duplicate real-time ticket alerts and memory leak after repeated tab switching	Medium	Socket.io event listeners were re-registered on every component render without teardown	Wrapped socket subscriptions in useEffect with dependency array and added socket.off() teardown handlers	Resolved	2026-09-25
BUG-008	Defect Liability Period (DLP) warranty countdown displayed "NaN Months Remaining" on certain browsers	Medium	Date parsing used browser-dependent Date.parse() which fails on non-standard date formats	Standardized all date storage to ISO 8601 (YYYY-MM-DD) and created a robust date-difference utility helper	Resolved	2026-09-26

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BUG-009	Marathi localization showed raw English keys for newly added hazard categories (e.g. DRAINAGE_OVERFLOW)	Medium	Dictionary divergence between en and mr translation objects in translations.ts	Synchronized all localization keys across English and Marathi dictionaries to achieve 100% bilingual parity	Resolved	2026-09-27
BUG-010	Mobile Citizen Feed navigation bar overlapped the bottom-most complaint cards on small screens	Low	Global layout container lacked viewport bottom safe-area offset padding when fixed navigation was active	Added dynamic viewport padding pb-24 to main content wrappers in BottomMobileNav.tsx and App.tsx	Resolved	2026-09-28
BUG-011	KPI banner showed hardcoded counts that did not update when new tickets were filed	Medium	Dashboard component used static mock data rather than deriving counts dynamically from live tickets state	Refactored KpiBanner.tsx to compute active grievances and SLAs dynamically from live database state	Resolved	2026-09-29
BUG-012	AI Chatbot suggestion chips did not autofill the complaint modal when clicked	Low	Chatbot modal was not communicating state to ComplaintFormModal due to missing parent callback handler	Passed shared callback onSelectHazard(category title) from parent App.tsx to enable 1-click complaint pre-filling	Resolved	2026-09-29
BUG-013	Dark mode text contrast was illegible when printing reports or viewing on low-contrast projectors	Low	CSS variables for background and text lacked high-contrast fallback tokens for print stylesheets	Added explicit @media print CSS overrides and high-contrast color classes for municipal report exports	Resolved	2026-09-30
BUG-014	GPS coordinate geofence allowed coordinates outside Nashik Municipal Corporation boundaries	High	Absence of geographic bounding box verification allowed test coordinates from Mumbai into Nashik wards	Implemented bounding box filter (lat: 19.85 - 20.15 lng: 73.65 - 73.95) rejecting coordinates outside NMC limits	Resolved	2026-10-01